Chapter 10 • Graphs of Functions

Solved Review Exercise 10

MCQs

1. x=5 represents:
a) x-axis
b) y-axis
c) line parallel to x-axis
d) line parallel to y-axis
Ans: d
A constant-x equation is a vertical line.

2. Slope of y=5x+3 is:
a) 3
b) −3
c) 5
d) −5
Ans: c
In y=mx+c, m=5.

3. The y-intercept of y=−2x−1 is:
a) −2
b) 2
c) −1
d) 1
Ans: c
At x=0, y=−1.

4. The graph of $y=x^3$ cuts the x-axis at:
a) x=0
b) x=1
c) x=−1
d) x=2
Ans: a
$x^3=0$ gives x=0.

5. The graph of 3^x represents:
a) growth
b) decay
c) both growth and decay
d) a line
Ans: a
Base 3 is greater than 1.

6. The graph of $y=-x^2+5$ opens:
a) upward
b) downward
c) left side
d) right side
Ans: b
The coefficient of x^2 is negative.

7. The graph of $y=x^2-9$ opens:
a) upward
b) downward
c) left side
d) right side
Ans: a
The coefficient of x^2 is positive.

8. $y=5^x$ is which function?
a) linear
b) quadratic
c) cubic
d) exponential
Ans: d
The variable is in the exponent.

9. Which is a reciprocal function?
a) $y=7^x$
b) $y=\frac{2}{x}$

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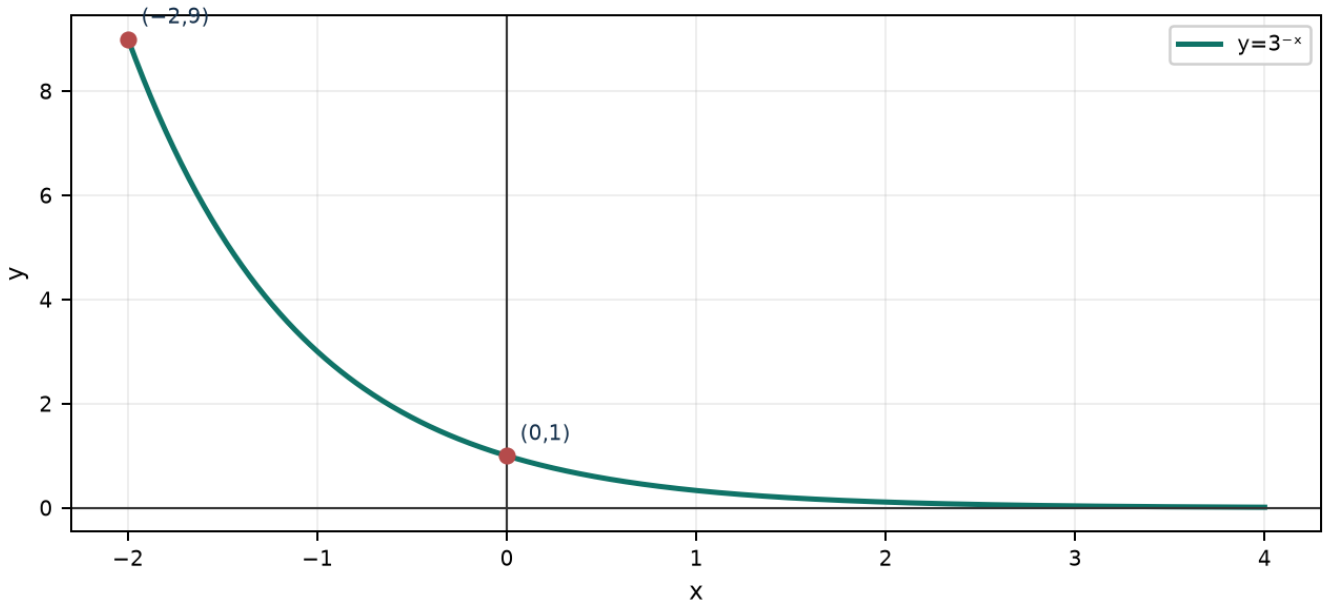
c) $y=2x^2$
d) $y=5x^3$
Ans: b
It has the variable in the denominator.

10. $y=-3x^3+7$ is which function?
a) exponential
b) cubic
c) linear
d) reciprocal
Ans: b
Its highest power is x^3.

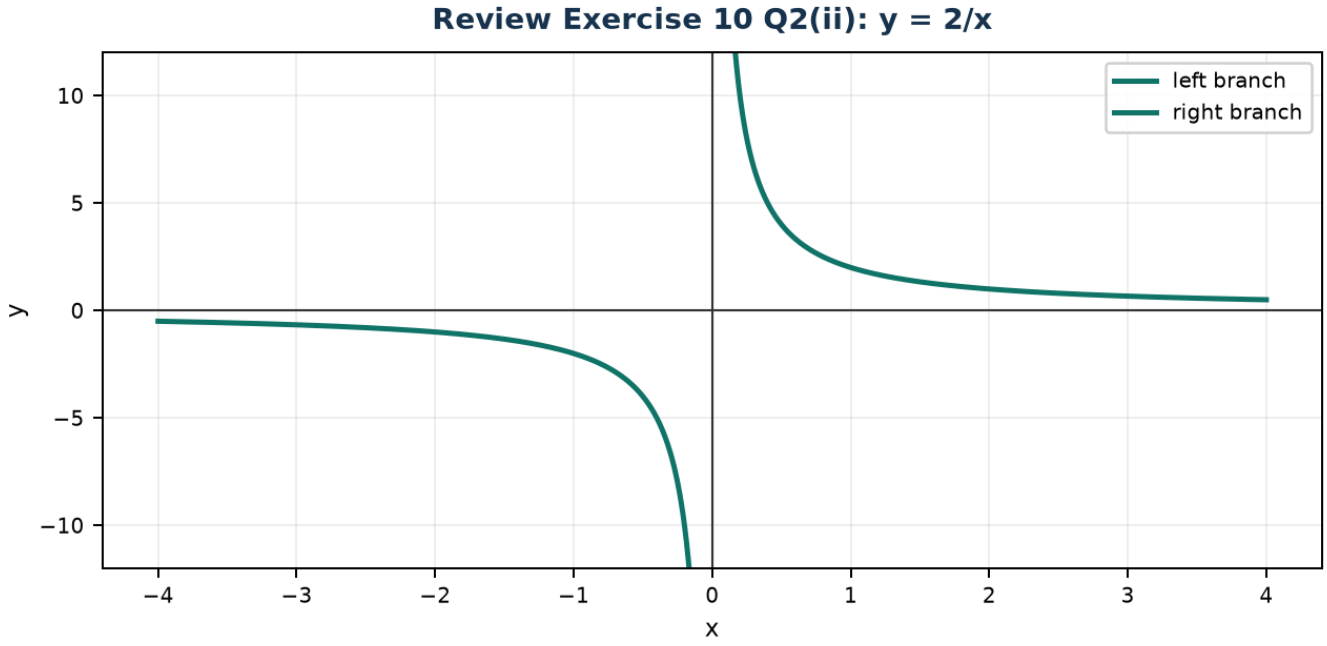
x	$y=3^{-x}$
−2	9
−1	3
0	1
1	1/3
2	1/9
3	1/27
4	1/81

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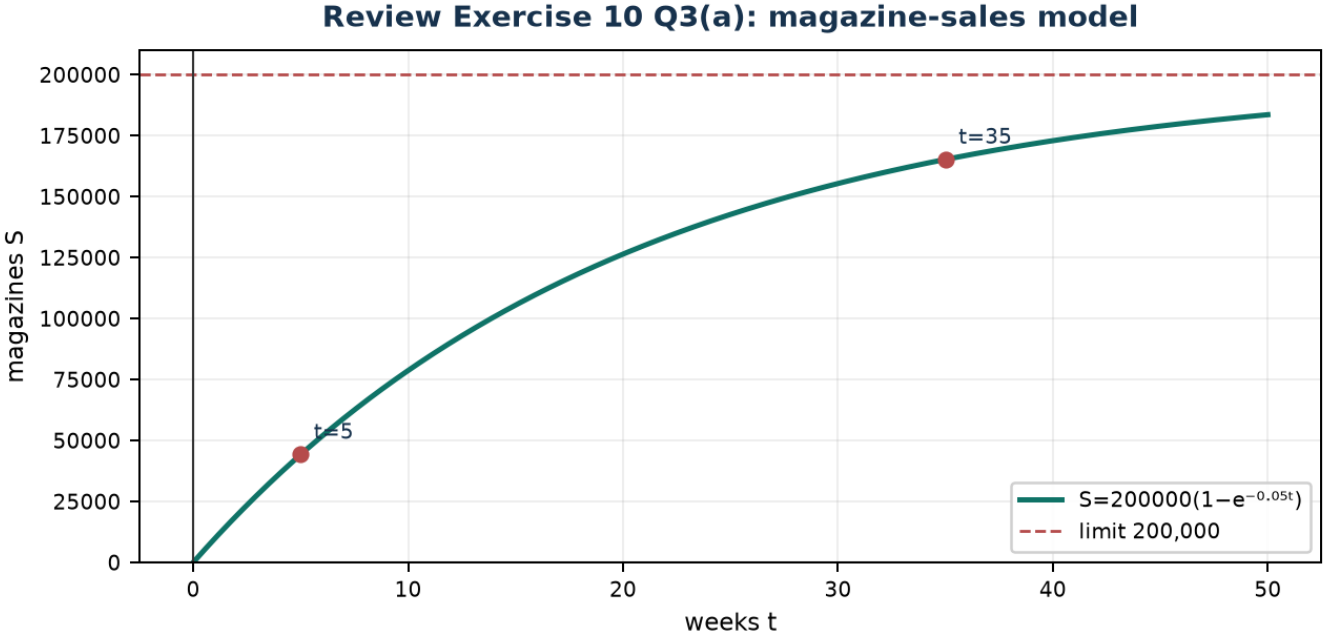
Review Exercise 10 Q2(i): $y = 3^{-x}$



x	y=2/x
−4	−1/2
−2	−1
−1	−2
1	2
2	1
4	1/2



weeks t	magazines S
0	0
5	44,239.8434
10	78,693.8681
20	126,424.1118
35	165,245.2113
50	183,583.0003



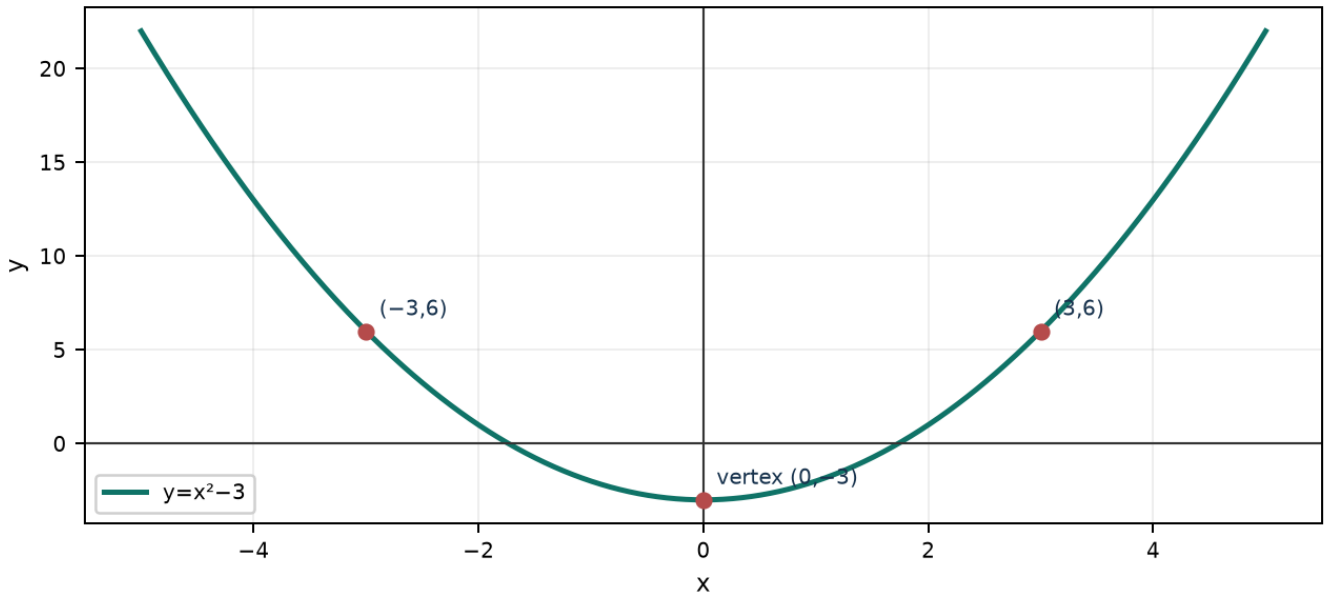
3(b). Using $S=200000(1-e^{-0.05t})$, calculate the number of magazines sold when t=5 and t=35.

At t = 5: S

$= 200000(1-e^{-0.25})$
$= 44,239.8434...$
At t = 35: S
$= 200000(1-e^{-1.75})$
$= 165,245.2113...$
Round only after evaluating the exponential expressions.
Approximately 44,240 magazines at 5 weeks and 165,245 at 35 weeks.

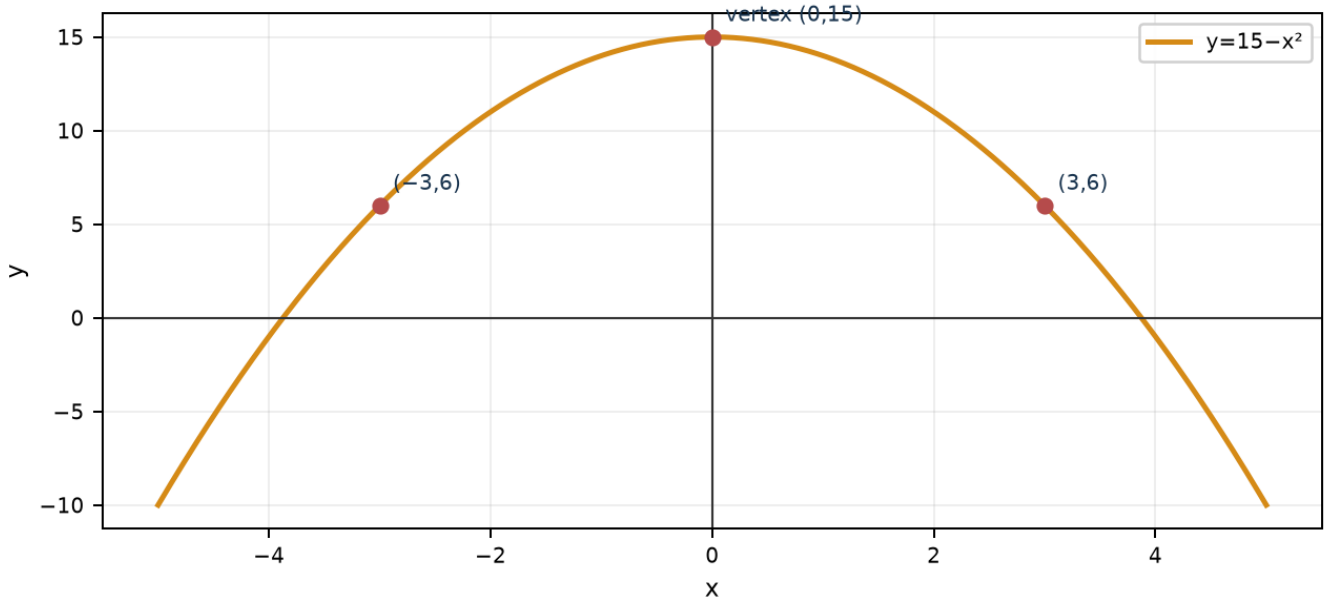
x	$y=x^2-3$
-5	22
-3	6
-2	1
0	-3
2	1
3	6
5	22

Review Exercise 10 Q4(i): y = x² − 3



x	$y=15-x^2$
-5	-10
-3	6
-2	11
0	15
2	11
3	6
5	-10

Review Exercise 10 Q4(ii): y = 15 − x²



x	y=x²−3	y=15−x²
-5	22	-10
-3	6	6
0	-3	15
3	6	6
5	22	-10

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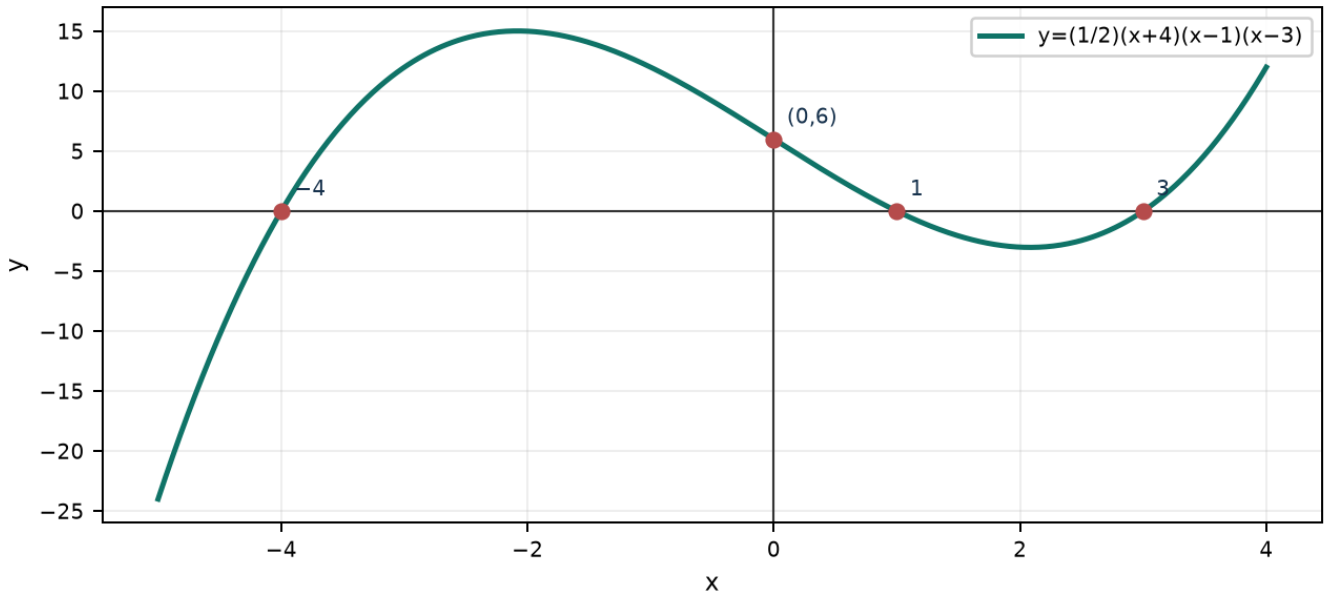
Review Exercise 10 Q4: intersecting parabolas



x	y
-5	-24
-4	0

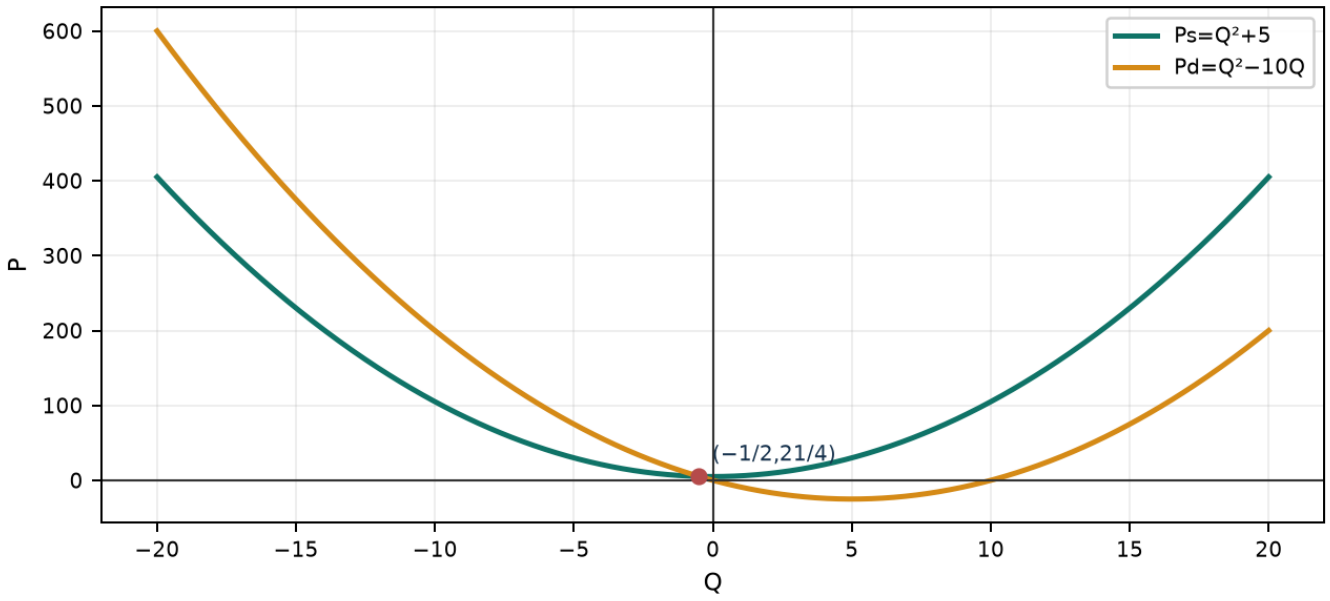
x	y
-2	15
0	6
1	0
2	-3
3	0
4	12

Review Exercise 10 Q5: cubic graph



Q	Ps	Pd
-20	405	600
-0.5	5.25	5.25
0	5	0
5	30	-25
10	105	0
20	405	200

Review Exercise 10 Q6: printed supply and demand curves



LEDs x	C(x) (Rs.)	R(x) (Rs.)
0	60,000	0
50	72,500	60,000
63.1579	75,789.4737	75,789.4737
64	76,000	76,800
100	85,000	120,000

Review Exercise 10 Q7: LED break-even model

